

- 2 (a) State Newton's Second Law of motion.

The rate of change in the momentum of a body is proportional to the resultant external force that acts on it, and the change in momentum is in the direction of the force.

[1]

- (b) A jet of water hits a vertical wall at right angles, as shown in Fig. 2.1. The jet of water has density ρ , cross-sectional area A , and hits the vertical wall with impact velocity u . The water then runs down the wall after impact with the wall.

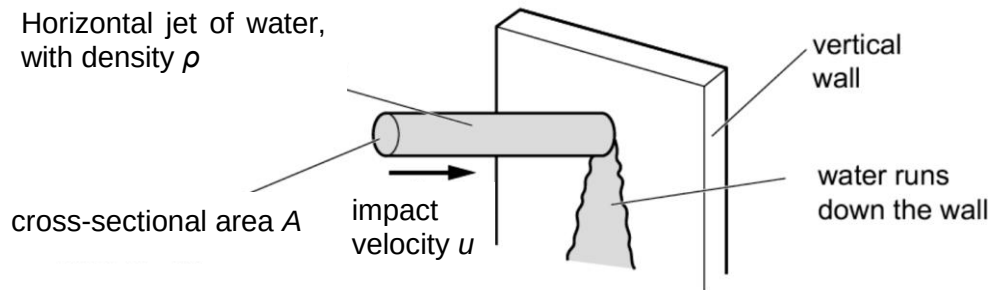


Fig. 2.1

- (i) Using Newton's Law of motion, show that the magnitude of the average force exerted on the water by the wall is

$$F = \rho Au^2.$$

From Newton's Second Law,

Force on water, F = rate of change of momentum of water

$$\begin{aligned} \text{Average } F &= \frac{\Delta p}{\Delta t} \\ &= \frac{M(0 - u)}{\Delta t} \\ &= -\left(\rho \times \frac{V}{\Delta t}\right)u = -\left(\rho \times \frac{Al}{\Delta t}\right)u = -\rho(Au)u \\ &= -\rho Au^2 \end{aligned}$$

Hence, magnitude of $F = \rho Au^2$

Equation for N2L [1]

Sub in final velocity = 0 [1]

Sub in $(M/t) = \rho Au$ [1]

[3]

- (ii) The density of water ρ is 1000 kg m^{-3} . Given that the jet of water in (b) has cross-sectional area A of 1.5 cm^2 and impact velocity u of 5.0 m s^{-1} ,

1. Calculate the magnitude of the average force exerted on the wall by the water. Explain your answer.

Force by wall on water = $\rho Au^2 = (1000)(1.5 \times 10^{-4})(5.0^2) = 3.75 = 3.8 \text{ N}$ [1]

By Newton's 3rd Law, magnitude of force by water on wall is equal to that by wall on water [1]

Hence magnitude of average force by water on wall = 3.8 N

magnitude of average force = N [2]

2. On Fig 2.2, sketch a graph to show the variation of pressure p on the wall with impact velocity u . [1]

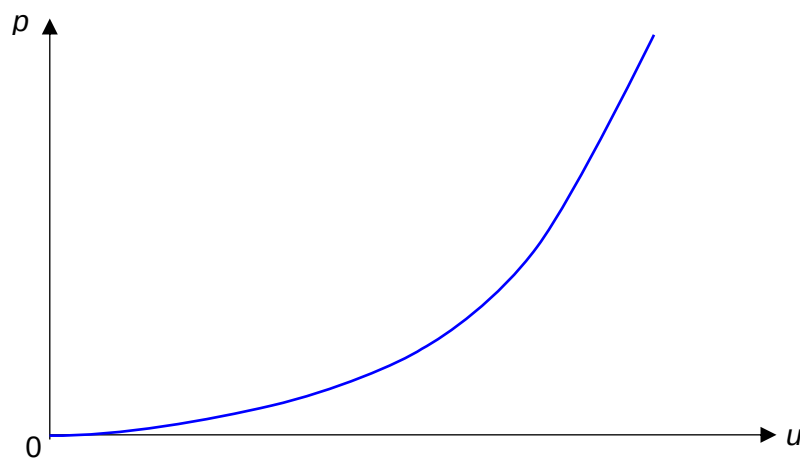


Fig. 2.2

3. Suggest the change, if any, to pressure P if the cross-sectional area A of the water jet is doubled.

No change.

(Since Pressure = Force/Area = $F/A = (\rho Au^2)/A = \rho u^2$, pressure is independent of A)

[Total: 8]

2. A ball of mass M of 750 g is held on a smooth horizontal surface between two identical springs